X100 CLEAN-SHEET | PHYSICAL SCHEMATIC

VBAT 3S-6S -> 5V_MAIN -> 3V3_MCU / 3V3_SENSOR / protected EXT_5V

USB: DATA ONLY + VBUS SENSE. USB_VBUS never joins 5V_MAIN.

Physical pins are wired in subsheets. Only nonblocking board/test qualifications are listed in GATE1 exports.

01_POWER

VBAT_RAW
5V_MAIN
3V3_SENSOR
GND
3V3_MCU
EXT_5V

PHYSICAL SUBSHEET
PIN-NET TABLE EXPORTED

02_MCU

3V3_MCU
SPI1_MISO
CAN_RX
TELEM_RX
VBAT_ADC
SWCLK
SPI1_MOSI
BMI088_GYRO_CS
GPS_TX
RC_TX
OUT1
OUT3
OUT5
BARO_I2C2_SCL
USB_DP
SDMMC1_D0
SDMMC1_D2
SDMMC1_CMD
EXT_I2C1_SDA
NRST
GND
USB_VBUS_SENSE
GPS_RX
CAN_RX
RC_RX
CURRENT_ADC
SPI1_SCK
BMI088_ACC_CS
CAN_TX
TELEM_TX
SDMMC1_CK
OUT2
OUT4
OUT6
BARO_I2C2_SDA
USB_DM
SDMMC1_D1
SDMMC1_D3
EXT_I2C1_SCL
SWDIO

PHYSICAL SUBSHEET
PIN-NET TABLE EXPORTED

03_SENSORS

3V3_SENSOR
SPI1_SCK
BMI088_ACC_CS
SPI1_MISO
BARO_I2C2_SDA
GND
SPI1_MOSI
BMI088_GYRO_CS
BARO_I2C2_SCL

PHYSICAL SUBSHEET
PIN-NET TABLE EXPORTED

04_USB

GND
USB_DM
USB_DP
USB_VBUS_SENSE

PHYSICAL SUBSHEET
PIN-NET TABLE EXPORTED

05_SD

3V3_MCU
SDMMC1_CK
SDMMC1_D1
SDMMC1_D3
GND
SDMMC1_D0
SDMMC1_D2
SDMMC1_CMD

PHYSICAL SUBSHEET
PIN-NET TABLE EXPORTED

06_CAN

3V3_MCU
GND
CAN_RX
EXT_5V
CAN_TX

PHYSICAL SUBSHEET
PIN-NET TABLE EXPORTED

07_UART_IO

EXT_5V
GPS_TX
RC_TX
TELEM_RX
EXT_I2C1_SCL
GND
TELEM_TX
GPS_RX
RC_RX
EXT_I2C1_SDA

PHYSICAL SUBSHEET
PIN-NET TABLE EXPORTED

08_PWM_ADC_DEBUG

GND
OUT2
OUT5
OUT6
CURRENT_ADC
SWDIO
OUT1
OUT3
OUT5
VBAT_ADC
SWCLK
NRST

PHYSICAL SUBSHEET
PIN-NET TABLE EXPORTED

09 USB core and SD isolation — CANDIDATE

文件: 09_USB_CORE_POWER.kicad_sch

10 USB startup and current staging — CANDIDATE

文件: 10_USB_STARTUP_GUARD.kicad_sch

PHYSICAL SCHEMATIC; PCB HOLD: see local unresolved inputs
PHYSICAL SCHEMATIC; PCB HOLD: see local unresolved inputs

X100 Clean-Sheet

Sheet: /
File: X100_CleanSheet.kicad_sch

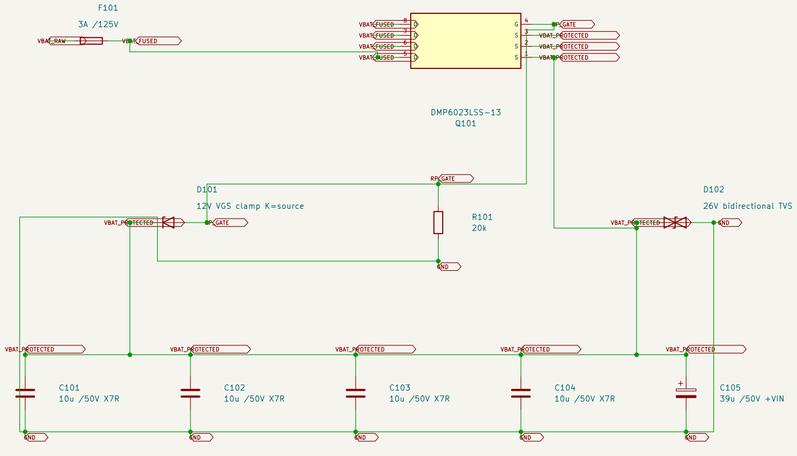
Title: X100 Clean-Sheet / hierarchy

Size: A2 | Date: 2026-10-04 | **Rev: Physical-1**
KiCad E.D.A. 10.0.6 | Id: 1/11

01 电源管理 / POWER MANAGEMENT

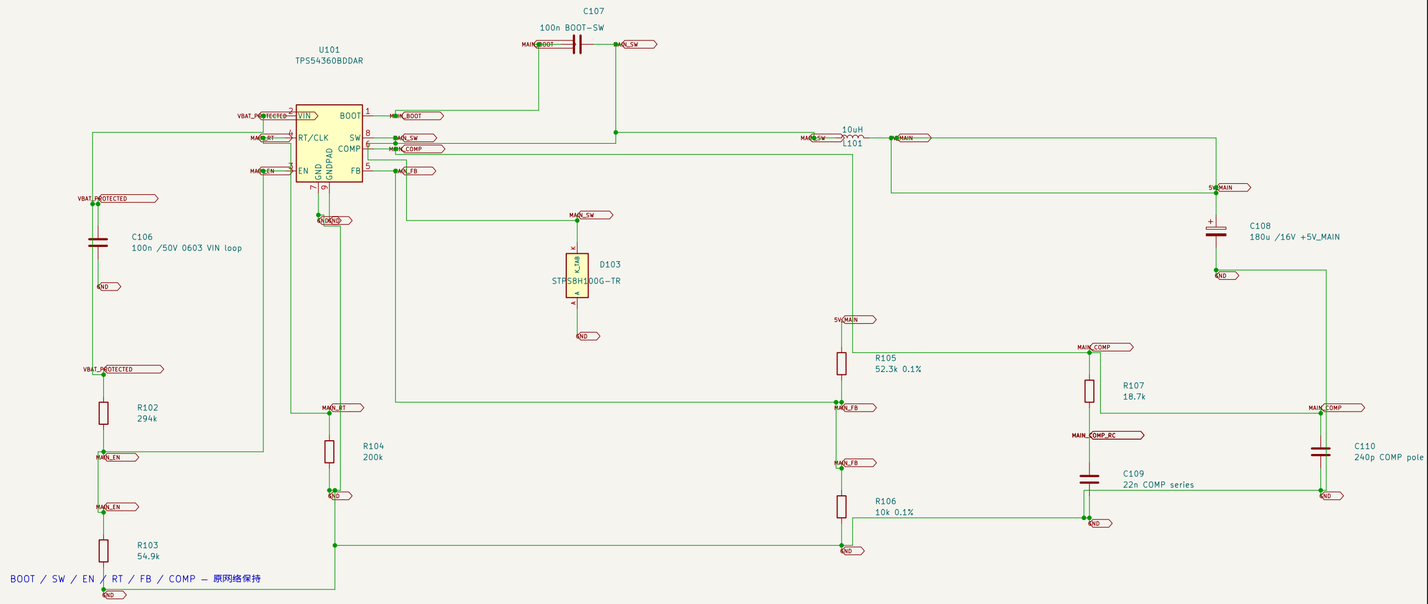
MAIN 5V: battery/bench only, 3S–6S. USB is never a supply.

VBAT 输入与保护 / VBAT INPUT AND PROTECTION

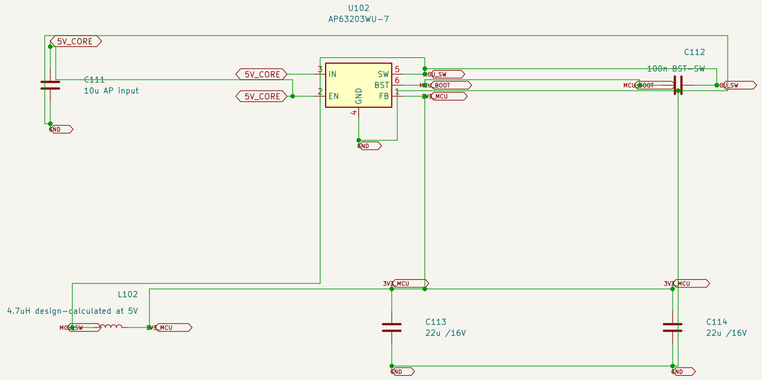


VBAT_RAW → VBAT_FUSED → VBAT_PROTECTED

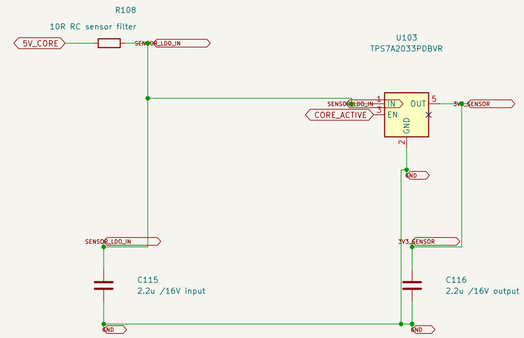
主 5V 降压电源 / MAIN 5V BUCK



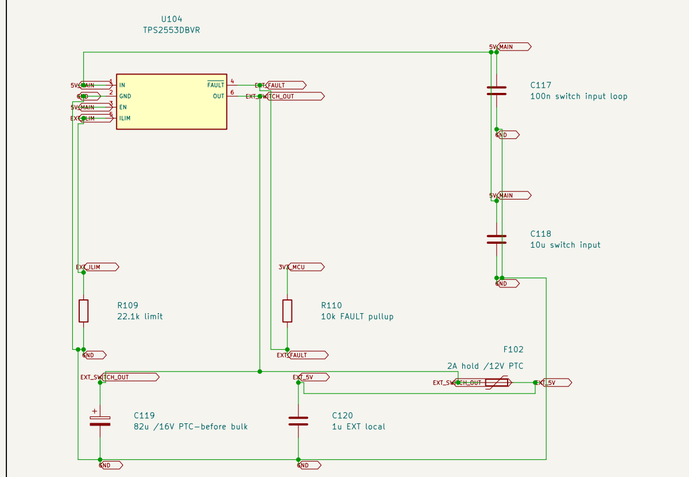
MCU 3.3V 电源 / MCU 3V3 SUPPLY



传感器 3.3V 电源 / SENSOR 3V3



受控外部 5V / CONTROLLED EXTERNAL 5V



VBAT / MAIN 5V

VBAT_RAW ↔ VBAT_RAW

5V_MAIN ↔ 5V_MAIN

MCU / SENSOR POWER

3V3_MCU ↔ 3V3_MCU

3V3_SENSOR ↔ 3V3_SENSOR

EXTERNAL 5V / GND

GND ↔ GND

EXT_5V ↔ EXT_5V

TP101 VBAT_RAW

TP102 VBAT_FUSED

TP103 VBAT_PROTECTED

TP104 5V_MAIN

TP105 3V3_MCU

TP106 SENSOR_LDO_IN

TP107 3V3_SENSOR

TP108 EXT_SWITCH_OUT

TP109 EXT_5V

TP110 EXT_FAULT

TP111 GND

#FLG101 PWR_FLAG

#FLG102 PWR_FLAG

#FLG103 PWR_FLAG

#FLG104 PWR_FLAG

#FLG105 PWR_FLAG

#FLG106 PWR_FLAG

#FLG107 PWR_FLAG

#FLG108 PWR_FLAG

TPS2553 constant-current limiting; recovers after overload removal, subject to thermal protection.

See unresolved_items.md; local holds are not ERC failures

PCB_LAYOUT=HOLD; physical exact-part schematic

X100 Clean-Sheet

Sheet: /01_POWER/

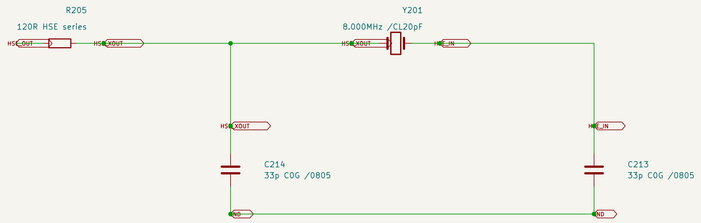
File: 01_POWER.kicad_sch

Title: 01_POWER

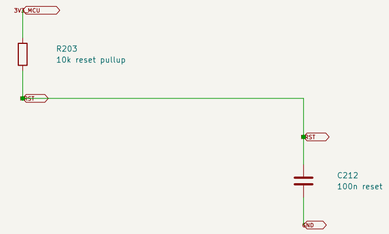
Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 2/11

02 主控与时钟 / MCU AND CLOCK

HSE 晶振 / HSE CLOCK · 8.000 MHz



复位 / RESET



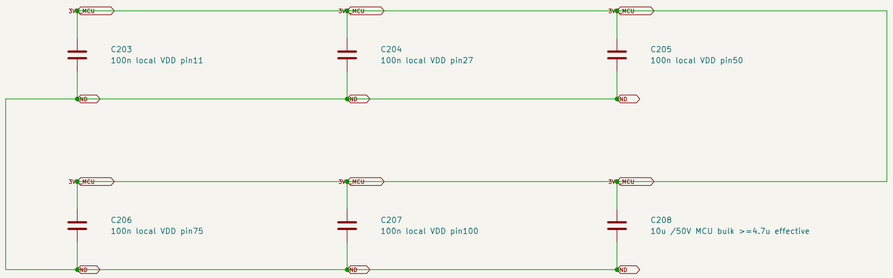
启动配置 / BOOT CONFIGURATION



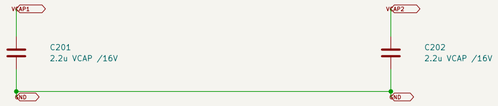
BOOT0 默认LOW · 恢复方式见页底说明

STM32H743 主控与去耦 / MCU CORE AND DECOUPLING

VDD 去耦 / VDD DECOUPLING



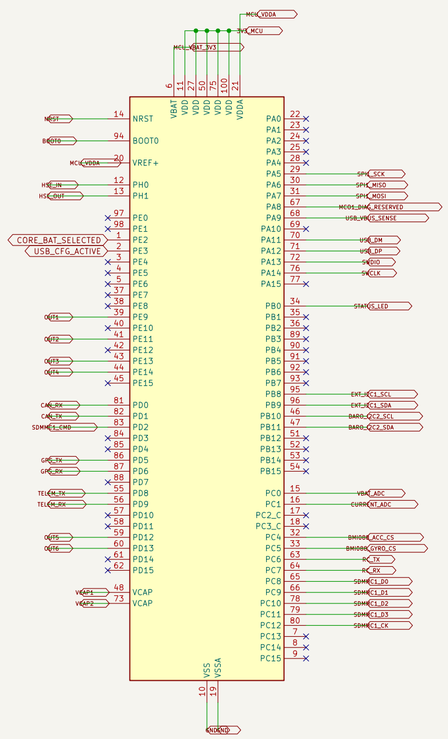
VCAP 核心稳压 / CORE REGULATOR



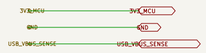




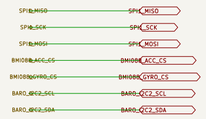
U201 STM32H743VIT6



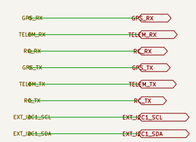
POWER / CONTROL



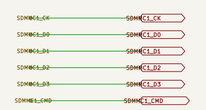
SPI / IMU / BARO



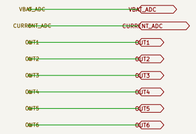
UART / I2C



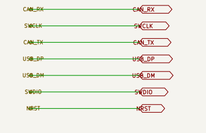
SDMMC



PWM / ADC

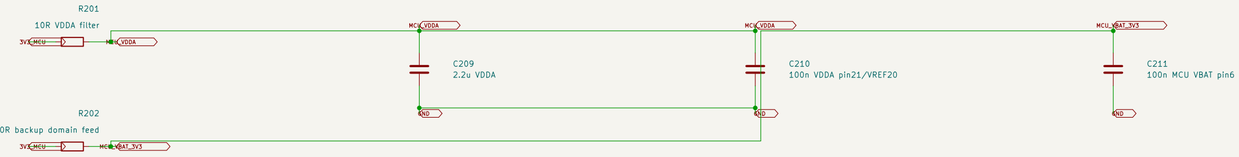


USB / CAN / DEBUG





模拟与备份供电 / ANALOG AND BACKUP POWER



状态与电源指示 / STATUS AND PWR LED



STM32H743VIT6 LQFP100. Frozen B allocation; unused GPIO has explicit NC.

BOOT0 recovery: strap TP_BOOT0 to TP_3V3 with power off; then reset. No flight jumper.

See unresolved_items.md; local holds are not ERC failures
PCB_LAYOUT=HOLD; physical exact-part schematic

X100 Clean-Sheet

Sheet: /02_MCU/
File: 02_MCU.kicad_sch

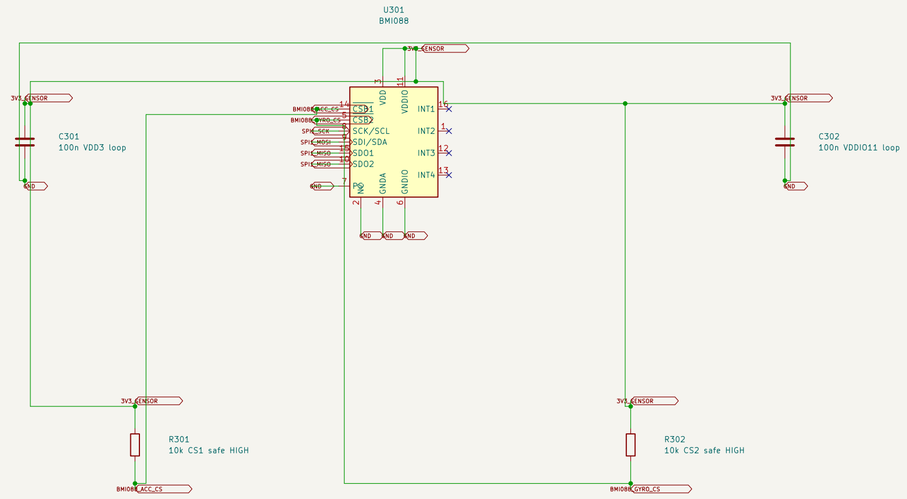
Title: 02_MCU

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 3/11

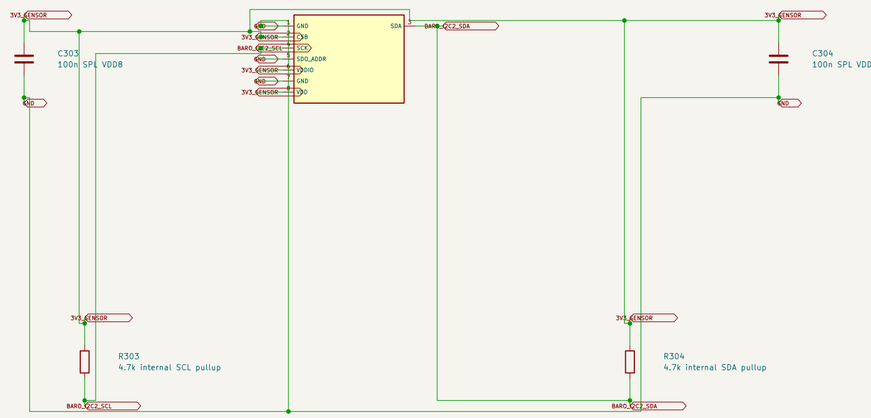
03 传感器 / SENSORS

Single BMI088 SPI1 and single SPL06-001 I2C2. No compass/second IMU.

BMI088 惯性测量 / BMI088 IMU · SPI1

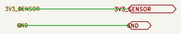


SPL06-001 气压计 / BAROMETER · I2C2

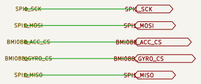


传感器接口与测点 / SENSOR INTERFACES AND TEST

POWER / GND

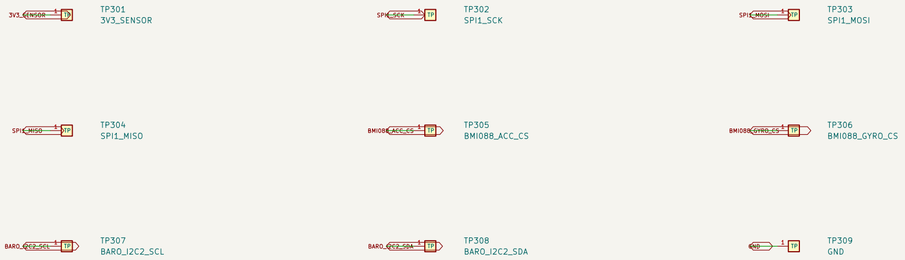


SPI1 / IMU



I2C2 / BARO





SPL: CSB2=VDDIO; SDO5=GND => I2C0x76. BMI: PS7=GND; NC2=GND.

SPI initialization must toggle accel CS and discard dummy read. Sensor-off GPIO must not drive.

See unresolved_items.md; local holds are not ERC failures
PCB_LAYOUT=HOLD; physical exact-part schematic

X100 Clean-Sheet

Sheet: /03_SENSORS/
File: 03_SENSORS.kicad_sch

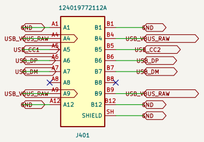
Title: 03_SENSORS

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 4/11

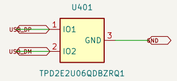
04 USB通信与保护 / USB INTERFACE AND PROTECTION

USB DATA ONLY + VBUS SENSE. Battery/bench power required.

USB Type-C 接口 / USB TYPE-C CONNECTOR



USB 数据线 ESD / DP-DM ESD



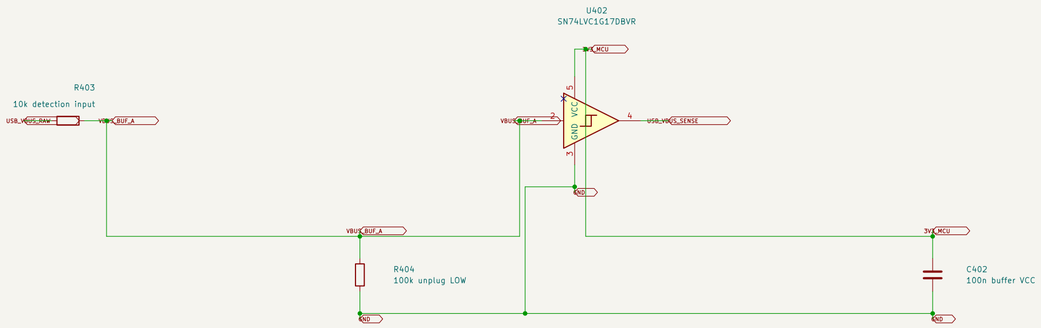
VBUS 保护 / USB VBUS PROTECTION



CC 配置与保护 / CC CONFIGURATION AND ESD



VBUS 检测 / VBUS SENSE



USB DP / DM



VBUS / POWER / GND





TP401
USB_VBUS_RAW



TP402
USB_VBUS_SENSE



TP403
USB_DP



TP404
USB_DM



TP405
GND

PA9 is software VBUS sense: configure GPIO input/session control; no raw5V on an unpowered MCU.

NO power OR, NO USB supply diode, NO USB connection to any main/sensor rail.

See unresolved_items.md; local holds are not ERC failures
PCB_LAYOUT=HOLD; physical exact-part schematic

X100 Clean-Sheet

Sheet: /04_USB/
File: 04_USB.kicad_sch

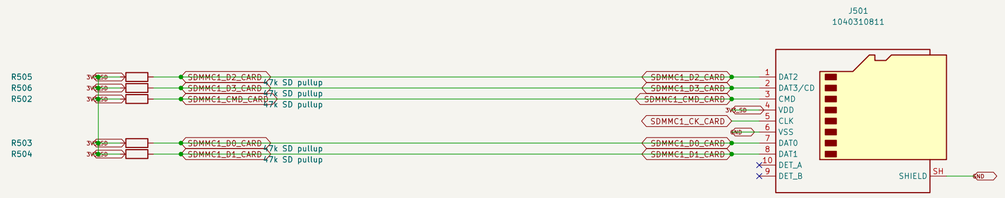
Title: 04_USB

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 5/11

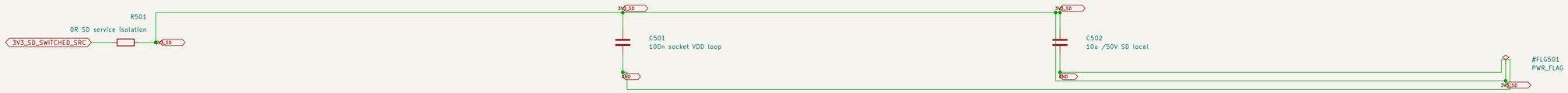
05 microSD数据存储 / MICROSD STORAGE

SDMMC1 4-bit, 3.3V only. Contact9/10 detect NC; no detect GPIO.

microSD 接口与信号上拉 / MICROSD AND SDMMC PULLUPS

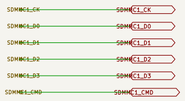


本地供电与去耦 / LOCAL POWER AND DECOUPLING



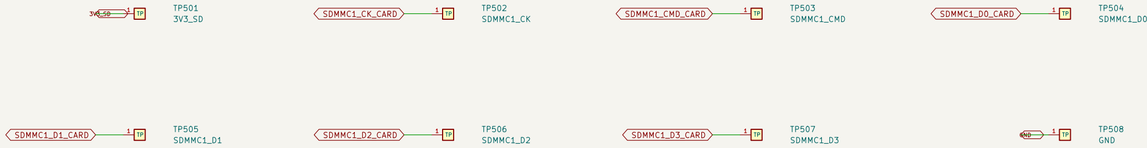
STM32 SDMMC1 ↔ microSD · 3.3V / 4-bit

SDMMC / SIGNALS



POWER / GND





No arbitrary damping resistor or 1.8V UHS. Series damping only after signal-integrity evidence.

See unresolved_items.md: local holds are not ERC failures
PCB_LAYOUT=HOLD: physical exact-part schematic

X100 Clean-Sheet

Sheet: /05_SD/
File: 05_SD.kicad_sch

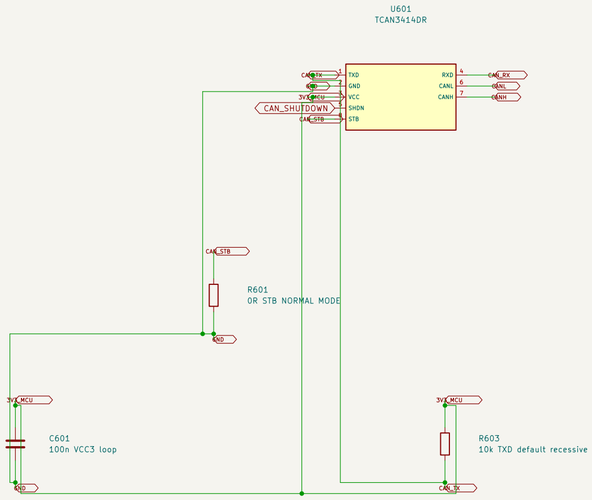
Title: 05_SD

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 6/11

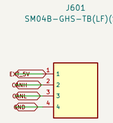
06 CAN通信 / CAN INTERFACE

TCAN3414DR 3.3V: Classical CAN only. SHDN=0,STB=0: Normal mode.

CAN 收发器 / CAN TRANSCEIVER



CAN 总线接口 / CAN BUS CONNECTOR



120Ω 终端 / CAN TERMINATION · 默认OPEN



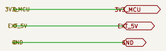
CANH → R602 → CAN_TERM → JP601 → CANL

按口与测点 / INTERFACES AND TEST POINTS

MCU / CONTROL



CANH / CANL / POWER













JP601 CAN TERM / 120R: manufacture OPEN (disabled). Manually solder bridge only at bus end (enabled). NO PASTE; wick clean to restore OPEN.

Built-in bus standoff +/-58V: receiver common mode +/-30V; bus-pin IEC ESD +/-10kV.

EXTERNAL_CAN_TVS=NOT_REQUIRED_FOR_V1. Whole-board IEC qualification deferred to test.

See unresolved_items.md: local holds are not ERC failures
PCB_LAYOUT=HOLD: physical exact-part schematic
X100 Clean-Sheet

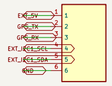
Sheet: /06_CAN/		
File: 06_CAN.kicad_sch		
Title: 06_CAN		
Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 7/11

07 串口与通用IO / UART AND GENERAL I/O

External5V is protected EXT_5V; all UART/I2C signals3.3V.

GPS 接口 / GPS INTERFACE

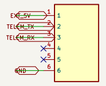
J701
SM06B-GHS-TB(LF)(SN)



GPS: FC_TX to receiverRX

遥测接口 / TELEMETRY INTERFACE

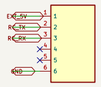
J702
SM06B-GHS-TB(LF)(SN)



TELEM:4/5 NC no flow control

RC 接收机 / RC RECEIVER · CRSF

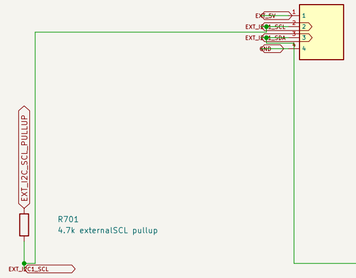
J703
SM06B-GHS-TB(LF)(SN)



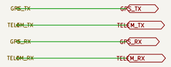
RC: CRSF3.3V logic:4/5 NC

外部 I2C / EXTERNAL I2C

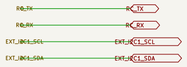
J704
SM04B-GHS-TB(LF)(SN)



GPS / TELEM

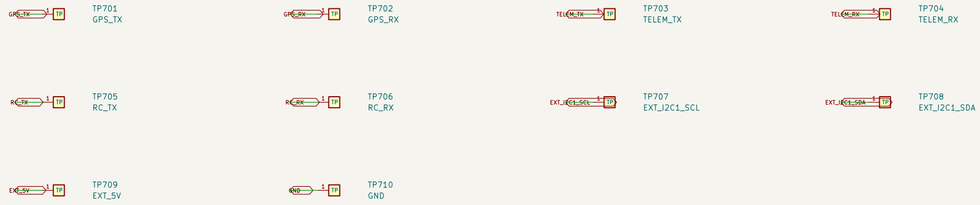


RC / EXTERNAL I2C



POWER / GND





GPS and externalI2C share I2C1. Combined pullups/cable capacitance must satisfy rise time.

See unresolved_items.md; local holds are not ERC failures
PCB_LAYOUT=HOLD: physical exact-part schematic
X100 Clean-Sheet
Sheet: /07_UART_IO/
File: 07_UART_IO.kicad_sch

Title: 07_UART_IO

Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 8/11

08 PWM / ADC / 调试 · PWM / ADC / DEBUG

PWM signal+GND only. Custom SWD5pin. Pack current NEVER passes through board.

六路 PWM 输出 / SIX PWM OUTPUTS

J801
OUT1 /GND



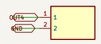
J802
OUT2 /GND



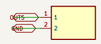
J803
OUT3 /GND



J804
OUT4 /GND



J805
OUT5 /GND

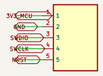


J806
OUT6 /GND



SWD 调试与电池输入 / SWD AND BATTERY INPUT

J807
SWD:1REF2GND3IO4CLK5NRST

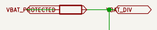


J808
BATTERY/bench3S–6S ONLY



电压采样 / VOLTAGE SENSE · ADC PC0

R801
150k 0.1%



R803
1k ADC series



VBAT_DIV

R802
10k 0.1% scale16

GND

VBAT_ADC_PRE_PROTECTION

C801
100n ADC LPF

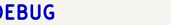
GND

电流输入采样 / CURRENT INPUT · ADC PC1

J809
CURRENT monitor only



R805
221k 1% >=100V



CURRENT_DIV

R806
20k 1% scale20/42.1

GND

R807
1k current limit

CURRENT_ADC_PRE_PROTECTION

C802
100n X7R 50V

GND

掉电隔离与保持 / POWERED–OFF ISOLATION

U801
TMUX1511PWR

3V3_MCU

C805
100n U801 VDD14

GND

3V3_MCU

R811
10k rail discharge; off bound 81.6mV

GND

ADC 输出滤波 / ADC OUTPUT FILTERS

R804
1k isolate ADC reservoir

VBAT_ADC

C803
22n X7R 50V ADC reservoir

GND

VBAT_ADC

R809
1.3M 1% discharge/IOFF bound

GND

R808
1k isolate ADC reservoir

1.3M 1% discharge/IOFF bound

CURRENT_ADC

C804
22n X7R 50V ADC reservoir

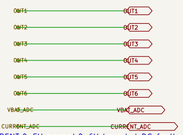
GND

CURRENT_ADC

R810

GND

PWM / ADC



DEBUG / POWER



CURRENT:0..5V normal,0..6V bounded DC fault, shared GND; no VBAT/negative-fault rating.

PC0/PC1 are FT_a. No rail-clamp diode added: external U801 powered-off isolation + bleed.

150k/10k remains nominal1/16; finite1.3M loading gives actual scale16.1401. Calibrate.

VBAT_DIV — TP801 VBAT_DIV

VBAT_ADC_PRE_PROTECTION — TP802 VBAT_ADC_PRE_PROTECTION

VBAT_ADC — TP803 VBAT_ADC

CURRENT_IN_RAW — TP804 CURRENT_IN_RAW

CURRENT_ADC — TP805 CURRENT_ADC

SWDIO — TP806 SWDIO

SWCLK — TP807 SWCLK

NRST — TP808 NRST

GND — TP809 GND

See unresolved_items.md: local holds are not ERC failures
PCB_LAYOUT=HOLD; physical exact-part schematic

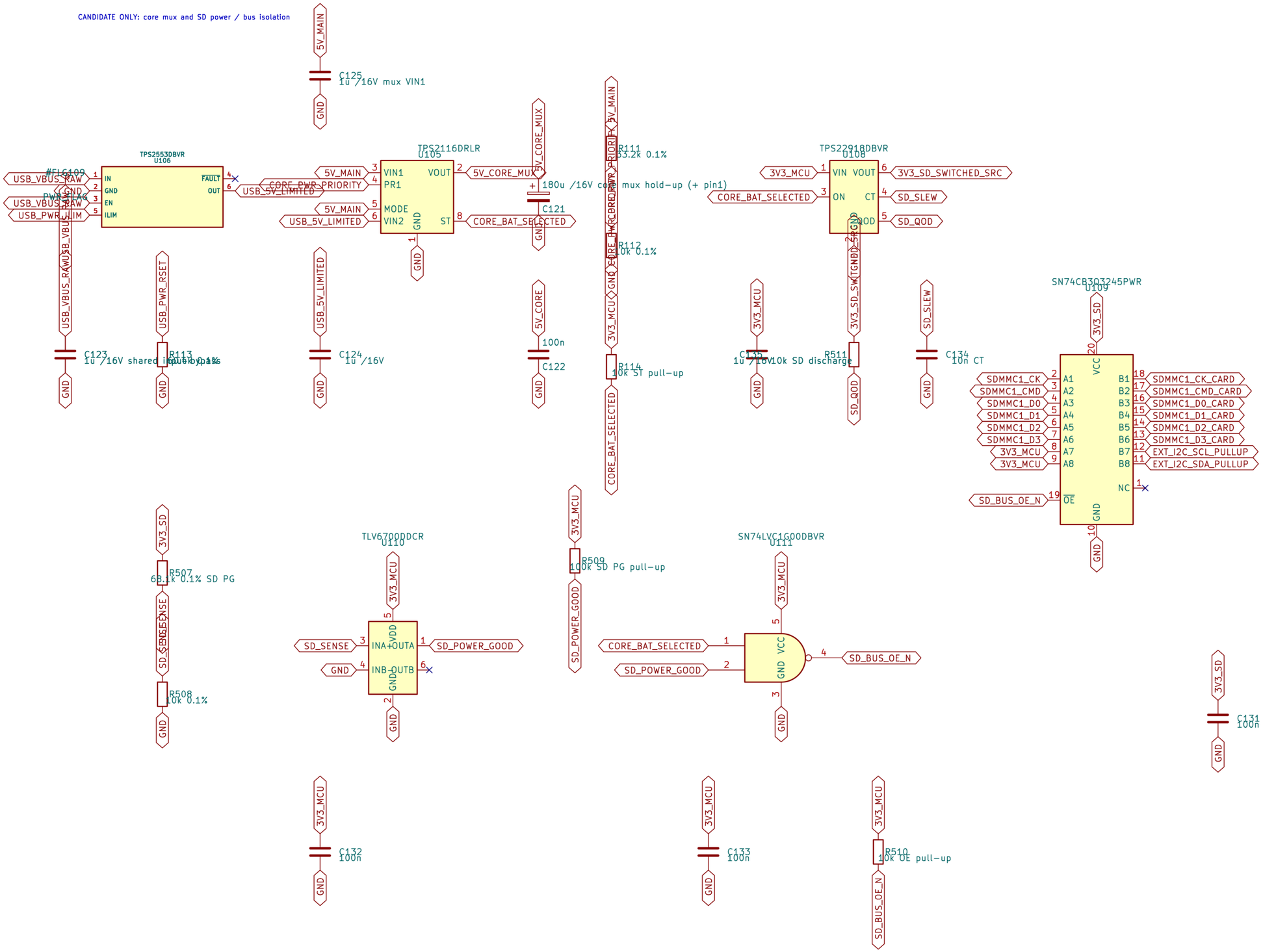
X100 Clean-Sheet

Sheet: /08_PWM_ADC_DEBUG/
File: 08_PWM_ADC_DEBUG.kicad_sch

Title: 08_PWM_ADC_DEBUG

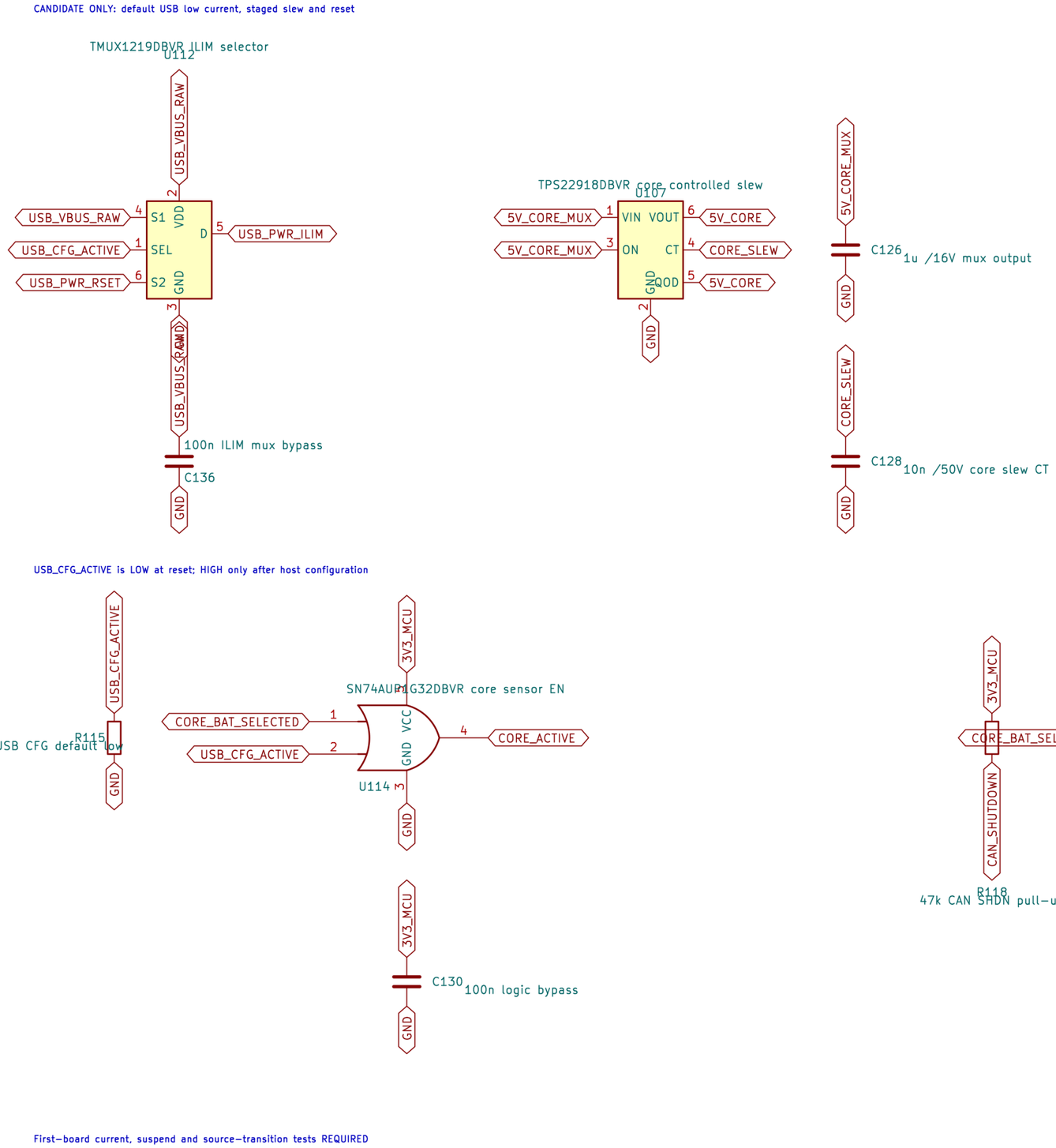
Size: A2	Date: 2026-10-04	**Rev: Physical-1**
KiCad E.D.A. 10.0.6		Id: 9/11

09 USB核心供电候选（源电路不变，仅扩大导出视窗）



原始图元贴近纸张边界；本页以扩大视窗的SVG完整显示。候选方案尚未获得整板电源/功能验收。

10 USB分阶段启动候选（源电路不变，仅扩大导出视窗）



原始图元贴近纸张边界；本页以扩大视窗的SVG完整显示。候选方案尚未获得整板电源/功能验收。